Materials

Your TA will lend your group:

- A 3D printed model of the graph $z = f(x, y)$ of some real-valued function f.
- A wax craft stick.

Please do NOT draw on the 3D printed model and return the materials to your TA at the end of tutorial.

Exercises

Orientation: Assume the real-valued function $f : D \to \mathbb{R}$ has domain $D = [0, 6] \times [0, 6]$. The highest corner of your 3D model corresponds to the point $(0, 0) \in D$ and the lowest corner corresponds to the point $(6, 0) \in D$.

T10.1 Define $g : \mathbb{R}^2 \to \mathbb{R}$ by $g(x, y) = (x - 2.5)^2 + (y - 2.5)^2$ and the circle given by

$$S = \{(x, y) \in \mathbb{R}^2 : g(x, y) = (1.5)^2\}.$$

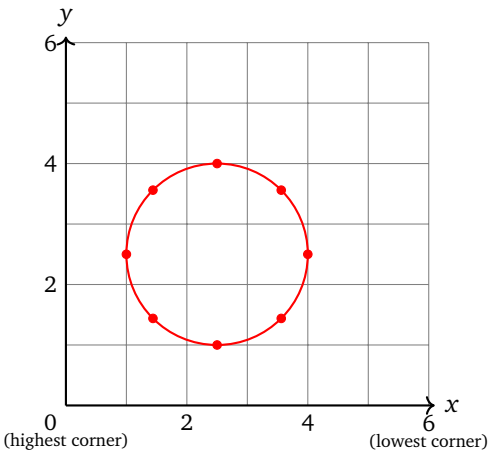
Affix the wax craft stick onto the model so that it corresponds to the constraint S.

(a) The set S is a subset of $\mathbb{R}^2$, so the wax craft stick is not exactly a representation of S itself. What set $T \subseteq \mathbb{R}^3$ does it instead represent? Express T with set builder notation.

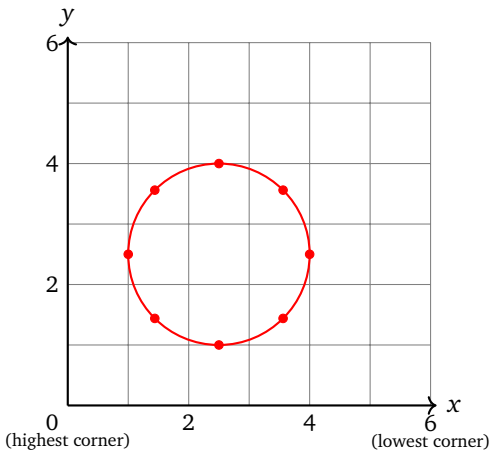
(b) Where are the local extrema of f on S? Write down the approximate coordinates.

(c) Use your 3D model to sketch the gradient vectors at each of the given 8 points along S.

$\nabla f(x, y)$ along S.

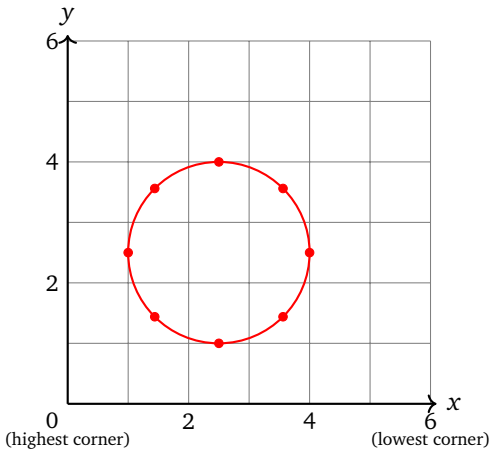


$\nabla g(x, y)$ along S.



Once finished, check your sketch of $\nabla f(x, y)$ with this Math3D demo.

(d) Imagine that the model is a mountain and the wax craft stick is a hiking trail loop. You are hiking along the trail. At a given point, if you want to climb higher along the trail, then should you travel clockwise (**cw**) or counterclockwise (**ccw**)? Use your 3D model to decide. Label each point with your choice (**cw** or **ccw**).



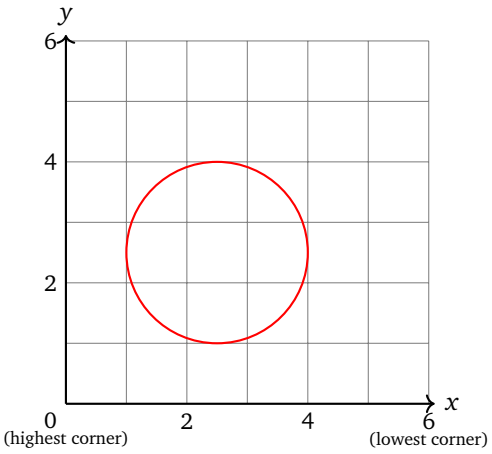
(e) Explain how your sketch from (c) also informs which way you should travel in (d).

(f) Find a point $(a, b) \in S$ and a point $(c, d) \in S$ such that:

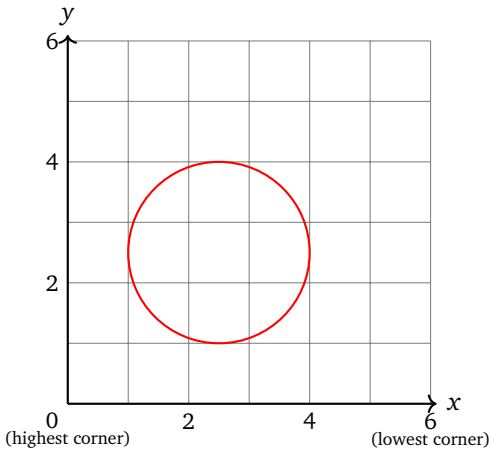
- the level set of f containing (a, b) is "**crossing**" the constraint S at (a, b).
- the level set of f containing (c, d) is "**tangent**" to the constraint S at (c, d).

Label each point and, using your 3D model, sketch the corresponding level set of f below.

$\{(x, y) \in D : f(x, y) = f(a, b)\}$



$\{(x, y) \in D : f(x, y) = f(c, d)\}$



Once finished, check that your level set drawings are accurate with this Math3D demo.

(g) On the corresponding diagrams, sketch the gradient vectors of f and g at the intersecting point on S, i.e. (a, b) for the lefthand diagram and (c, d) for the righthand diagram. How do they compare in each picture?

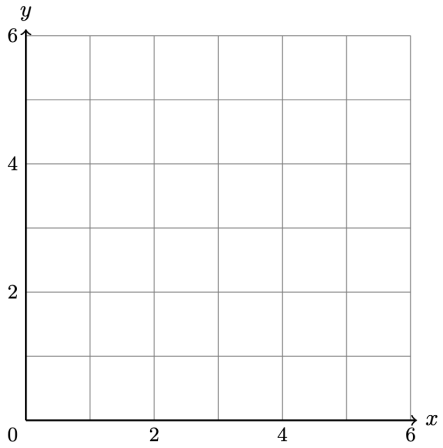
T10.2 Continue to use your model for the following questions, but remove your wax craft stick. You will be using the craft stick to visualize other constraint curves. Be creative! There are multiple correct answers.

(a) Using the wax craft stick, can you find a "closed curve" constraint $S = \{(x,y) \in (0,6)^2 : g(x,y) = 0\}$ for some new function $g : \mathbb{R}^2 \to \mathbb{R}$ that has a specified number of solutions to the Lagrange multiplier system?

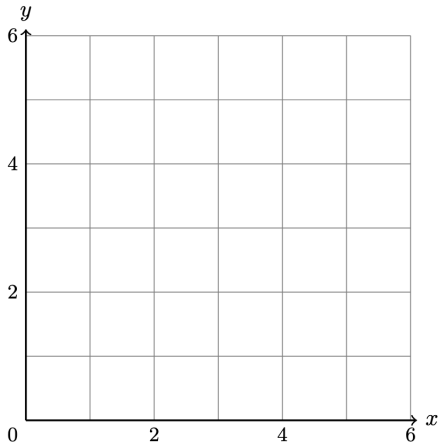
$$\nabla f(x,y) = \lambda \nabla g(x,y), \qquad g(x,y) = 0$$

Draw your proposed constraint and label the solutions on the plots below. (No formula required for g.)

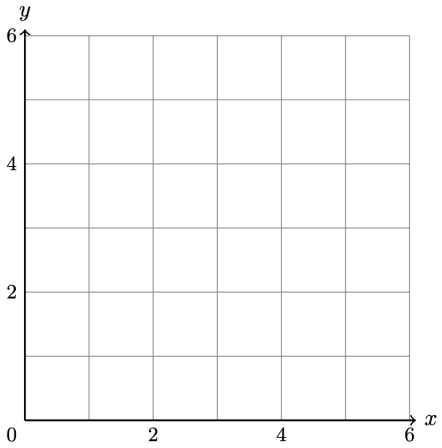
Exactly 2 solutions.



Exactly 4 solutions.



∞ many solutions.

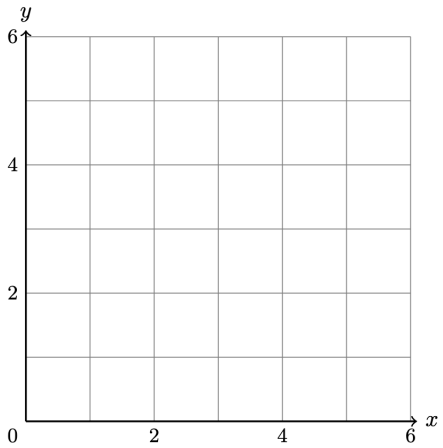


(b) Can you place a "closed curve" constraint on the interior $(0,6)^2$ with 1 or fewer solutions to the Lagrange multiplier system? Explain why or why not.

(c) Consider the diagonal line constraint $S = \{(x,y) \in [0,6]^2 : y = x\}$.

i) How many local extrema of f on S are there?
ii) How many solutions to the Lagrange multiplier system will there be?

Sketch and label your answers on the plot below. *Hint:* Your answers to i) and ii) should differ.



Once finished, check your solutions with this Math3D demo.